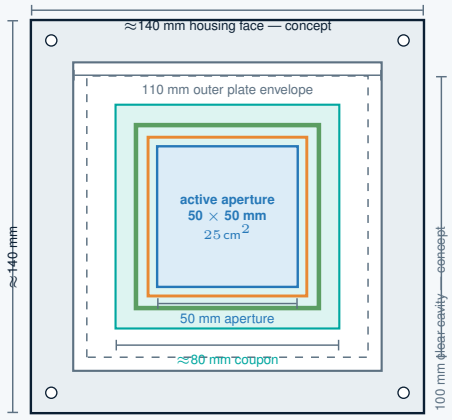
Integrated EO two-chamber cell — dimensioned concept views

Current sealed two-chamber system: governing values, starting values and fabrication-release dimensions to close.

A Chamber face and central membrane cartridge

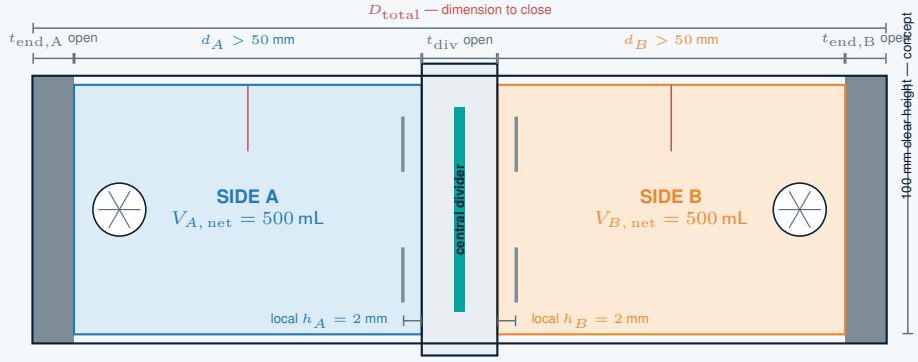


Nested face dimensions
Housing face: ≈140 × 140 mm [concept]
Clear chamber face: 100 × 100 mm [concept]
Outer plate envelope: 110 × 110 mm [starting]
Membrane coupon: ≈80 × 80 mm [starting]
Active aperture: 50 × 50 mm [fixed]

Radial lands on every edge
Housing edge → outer plate: 15 mm [concept]
Outer plate → coupon edge: 15 mm [starting]
Coupon edge → aperture: 15 mm [starting]

Within the 15 mm coupon land
3 mm inner seal + 2 mm vented guard groove +
3 mm outer seal + 7 mm spare land [starting]

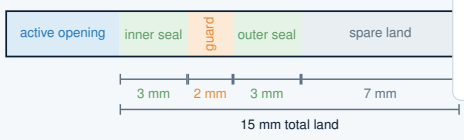
B Axial section and depth dimensions to close



Volume rule: A 100 × 100 × 50 mm empty cavity is 500 mL gross. Set d_A and d_B from installed fan, probe, baffle and port displacement, then calibrate each assembled chamber to 500 mL net free gas. Gross depth is a dimension to close.

C Seal-land chain and membrane-stack detail

Radial land chain: aperture → coupon



Sourced starting geometry
Maximum active aperture: 50 × 50 mm
Coupon: ≈80 × 80 mm
Outer plate envelope: 110 × 110 mm
Seal / guard / seal: 3 / 2 / 3 mm
Mask inserts: 40×40, 30×30, 20×20 and 8×50 mm

Stack through the active aperture — thickness exaggerated



Axial stack and local-gap status
Membrane thickness: 36.9 µm [model anchor]
Lot membrane thickness: open / measure
Porous-electrode thickness t_e: open / measure
Divider structure t_{div}: open / verify seal compression
Local membrane-face gap: 2 mm nominal [starting]
Interchangeable gap inserts: 1 / 2 / 4 mm
Active gap volume at 2 mm: 5.00 mL per side.

The local 2 mm membrane region contributes 5.00 mL to each calibrated 500 mL net free-gas inventory.

Starting bores: 2 per side, 6 mm ID × 8 mm long. Port threads, probes, fan, baffles and fasteners are dimensions to close.

Operating design state: 32 °C, initial $a_w = 0.50$, and membrane differential pressure target $|\Delta p| \leq 5$ Pa.

D Dimension and status register

Quantity	Value shown	Status
Net free gas volume	500 mL per side	Governing; calibrate
Chamber clear face	100 × 100 mm	Concept
Empty equivalent depth	50 mm per side	Gross-volume reference
Actual cavity depths	d_A, $d_B > 50$ mm	Dimension to close from displacement
Housing face	≈140 × 140 mm	Concept
Total axial depth	D_{total}	Dimension to close
Active aperture	50 × 50 mm = 25 cm²	Fixed design basis
Local membrane-face gap	2 mm nominal; 1 / 2 / 4 mm inserts	Sourced starting value
Active gap gas volume	5.00 mL per side at 2 mm	Subset of 500 mL net volume
Membrane coupon	≈80 × 80 mm	Sourced starting value
Outer plate envelope	110 × 110 mm	Sourced starting value
Inner seal / guard / outer seal	3 / 2 / 3 mm	Sourced starting value
Membrane thickness	36.9 µm	Model anchor; measure actual
Insulating frame thickness	8–10 mm	Starting value
Load-plate thickness	12 mm metal / ≥16 mm PEEK	Starting value; structural verification
End walls and divider	$t_{\text{end,A/B}}$, t_{div}	Dimension to close; structural verification
Port bores	2 × 6 mm ID × 8 mm long / side	Starting geometry
Probes, fans, baffles	envelope and location	Dimension to close
Fasteners and tolerances	pattern, size, flatness	Dimension to close

Fabrication-release basis: Fixed and starting values define the engineering concept. The 500 mL value is calibrated net free gas per side. Open items close through component selection, CAD displacement accounting, structural checks and calibration.

Engineering concept; fabrication drawing release follows closure of the dimension and tolerance register.